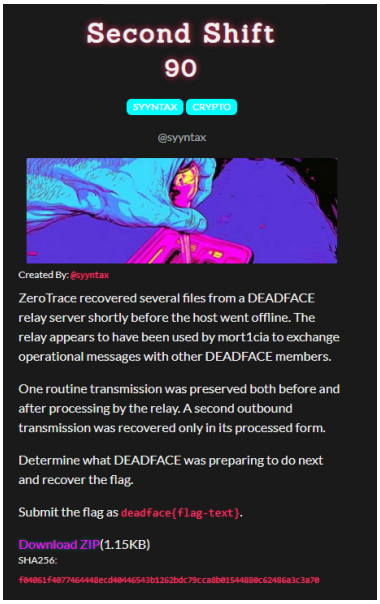
Solving “Second Shift” (90 pts - Cryptography)

A known-plaintext attack against reused stream-cipher/CTR keystream material.



Challenge page used during the solve.

1. The Challenge

The archive contained a plaintext heartbeat, an encrypted copy of that heartbeat, and a second encrypted tasking message. That is a strong hint that the same stream keystream or CTR nonce was reused.

2. Recover the Keystream

For a stream cipher, ciphertext = plaintext XOR keystream. Because the heartbeat plaintext and its encrypted form are both available, XORing them reveals the keystream byte-for-byte.

```
keystream = relay-heartbeat.json XOR relay-heartbeat.enc
```

3. Decrypt the Tasking

Apply the recovered keystream to relay-tasking.enc. The result is fully readable JSON.

```
{"type":"tasking","target":"aurora-vpn","action":"stage-access",
"window":"00:30Z","operator":"mort1cia",
"note":"Use the backup tunnel. Archive bundle under
DEADFACE{ctr_nonce_reuse_breaks_confidentiality} before exfil."}
```

4. Why the Attack Works

CTR mode is secure only when a nonce/counter stream is never reused with the same key. Reuse causes the same keystream to protect multiple messages, and one known plaintext exposes that keystream for the overlapping bytes.

5. The Flag

```
deadface{ctr_nonce_reuse_breaks_confidentiality}
```

6. General Method

• Look for plaintext/ciphertext pairs with similar sizes.

• XOR the known pair to obtain the stream.

• XOR the recovered stream with the second ciphertext.

• Validate by checking for structured plaintext such as JSON.

Summary

Field	Result
Category	Cryptography
Weakness	CTR/stream keystream reuse
Technique	Known-plaintext XOR
Flag	deadface{ctr_nonce_reuse_breaks_confidentiality}